

YASA: Band Power

This example demonstrates EEG band-power extraction using **YASA (bandpower)** with an MNE `Raw` object.

The workflow follows the EEG Feature Atlas structure:

EEG → Preparation → Method → Feature → Numerical Output → Visualization → Interpretation

We will:

1. download a public EEG recording;
2. load the EDF file with MNE;
3. standardize EEG channel names;
4. attach standard electrode positions;
5. calculate relative and absolute band power with YASA;
6. inspect the numerical output;
7. visualize average band power;
8. visualize spatial band-power patterns;
9. interpret the feature carefully;
10. summarize the feature for the Atlas.

Library / Method / Feature

Field	Value
Library	YASA
Function	<code>yasa.bandpower()</code>
Method	Welch power spectral density
Feature	EEG band power
Domain	Spectral + Spatial
Input	MNE <code>Raw</code>
Output	Channel × Band DataFrame
Visualization	Bar chart + scalp topomaps

Important distinction:

- Welch = spectral estimation method
- PSD = spectral representation

- Band power = feature extracted from the PSD
- Bar chart / Topomap = visualization

The domain is listed as **Spectral + Spatial** rather than Spectral alone: the feature itself (band power) is a spectral quantity, but this example also represents it spatially, one scalp topomap per band, the same reasoning used for the MNE PSD example earlier in the Atlas.

Dataset

We use the public **EEG Motor Movement/Imagery Dataset (EEGBCI)** distributed through PhysioNet.

The dataset contains 64-channel EEG recordings from multiple subjects and experimental runs.

Example recording:

- Subject: 1
- Run: 2
- Condition: baseline, eyes closed
- Recording type: continuous EEG

Run 2 corresponds to the eyes-closed baseline according to the MNE EEGBCI dataset documentation.

The data are downloaded automatically by MNE if they are not already available locally.

✓ **General**

Filename(s)	S001R02.edf
MNE object type	RawEDF
Measurement date	2009-08-12 at 16:15:00 UTC
Participant	X
Experimenter	Unknown

✓ **Acquisition**

Duration	00:01:01 (HH:MM:SS)
Sampling frequency	160.00 Hz
Time points	9,760

✓ **Channels**

EEG	64
Head & sensor digitization	67 points

✓ **Filters**

Highpass	0.00 Hz
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Recording information

Field	Value
Subject	1
Run	2
Channels	64
Sampling frequency	160.0 Hz
Duration	61.0 seconds

Preparation

The following preparation steps are explicit:

- EEGBCI channel-name standardization
- standard 10-05 montage assignment

No filtering, resampling, epoching, ICA, or artifact removal is performed.

This is intentional: the example focuses on the YASA feature rather than on a complete EEG preprocessing pipeline.

Band Power

Band power measures how much spectral power is contained within a predefined frequency range.

We use four commonly used EEG frequency bands:

- **Theta:** 4–8 Hz
- **Alpha:** 8–12 Hz
- **Beta:** 12–30 Hz
- **Gamma:** 30–45 Hz

YASA estimates the power spectrum using Welch's method and integrates power within the requested frequency bands.

With `relative=True`, YASA normalizes each band's power by the total power between the minimum and maximum frequencies specified by the requested bands.

For these bands, the normalization range is therefore:

4–45 Hz

This means relative power depends on the selected band definitions. Relative-power values should therefore not be compared directly with results obtained using a different normalization range.

YASA also provides total absolute power in the `TotalAbsPow` column. When `relative=False` is used, the returned band values represent non-normalized absolute power.

Feature parameters

The important parameters are specified explicitly rather than relying on hidden defaults:

- `relative=True` – return relative band power
- `win_sec=4` – 4-second Welch windows
- `bandpass=False` – do not apply an additional bandpass filter
- `average="median"` – Welch averaging method
- `window="hamming"` – Welch window function

This makes the feature extraction step easier to reproduce.

	Theta	Alpha	Beta	Gamma	TotalAbsPow	FreqRes	Relative
Chan							
FC5	0.308068	0.359125	0.267865	0.064943	829.872795	0.25	True
FC3	0.292130	0.395963	0.262167	0.049741	921.235923	0.25	True
FC1	0.303886	0.407151	0.247784	0.041179	983.025110	0.25	True
FCz	0.307745	0.396678	0.254622	0.040955	1021.435899	0.25	True
FC2	0.285799	0.407090	0.266850	0.040262	947.376044	0.25	True
...	...	...	...	...	...	...	...
PO8	0.065722	0.783663	0.136926	0.013690	4255.412407	0.25	True
O1	0.071272	0.764128	0.158007	0.006593	4433.074888	0.25	True
Oz	0.069420	0.735612	0.184306	0.010662	3586.640707	0.25	True
O2	0.064455	0.754053	0.170150	0.011342	4563.782291	0.25	True
Iz	0.111362	0.644952	0.218460	0.025226	2604.222821	0.25	True

64 rows × 7 columns

Numerical Output

The YASA result is a `pandas.DataFrame`.

- **Rows:** 64 EEG channels
- **Band-feature columns:** Theta, Alpha, Beta, Gamma
- **Additional output:** `TotalAbsPow`

Output structure:

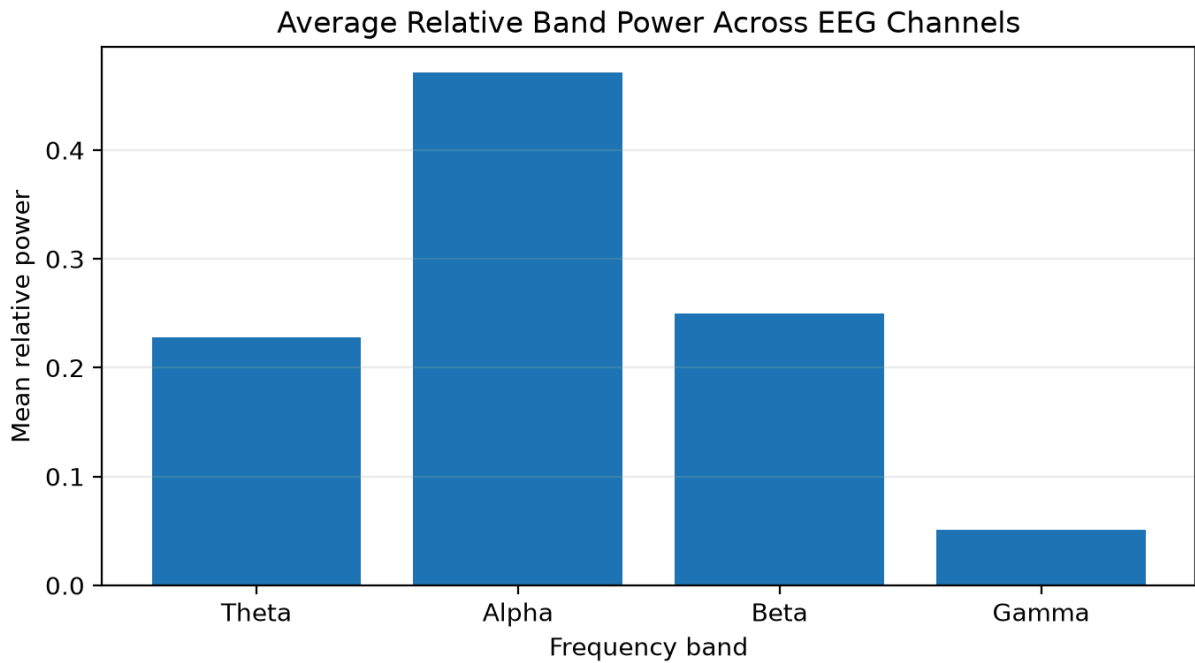
`Channel × Band`

Each row represents one EEG channel.

Each band column contains the relative band-power value calculated for that channel.

Available columns:

`Theta, Alpha, Beta, Gamma, TotalAbsPow, FreqRes, Relative`



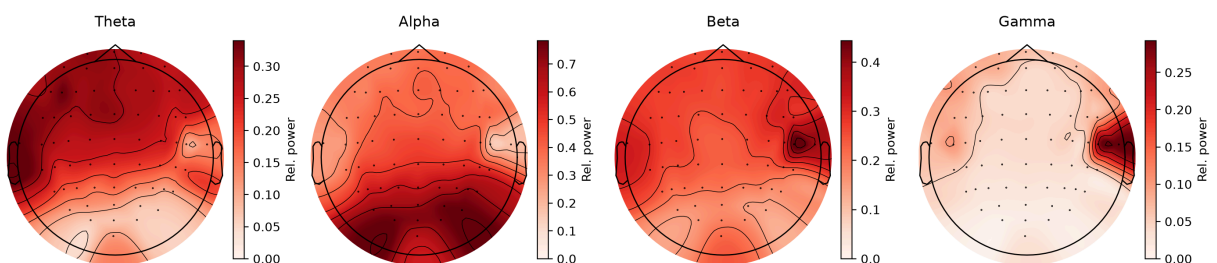
Spatial Visualization

Because the EEG channels have standard electrode positions, the channel-level band-power values can be visualized as scalp topomaps.

Each map shows the **spatial pattern within one frequency band**, with its own colorbar reporting relative power for that band.

The maps use independent color scales. Therefore:

- colors can be compared spatially within one map;
- colors should not be compared between different maps — always check each map's own colorbar;
- cross-band magnitude should be interpreted using the bar chart above.



Interpretation

Mathematical interpretation

Relative band power represents the proportion of spectral power assigned to a frequency band relative to the total power across the selected 4–45 Hz range.

Higher values therefore indicate that a larger proportion of the analyzed spectral power is located in that frequency band.

EEG interpretation

Band-power differences across channels describe differences in the spatial distribution of spectral power.

They should not automatically be interpreted as evidence of a specific cognitive, neurological, or clinical state.

In particular, the Gamma band (30–45 Hz) is more vulnerable to contamination from EMG, movement, electrode artifacts, and power-line noise. High Gamma power should therefore not be interpreted as cortical Gamma activity without appropriate artifact assessment.

This example does not perform ICA or other artifact-removal procedures.

Atlas Summary

Field	Value
Library	YASA
Method	Welch PSD
Feature	Band Power
Domain	Spectral + Spatial
Input	Continuous MNE <code>Raw</code>
Output	Channel × Band
Bands	Theta, Alpha, Beta, Gamma
Frequency range	4–45 Hz
Normalization	Relative power
Absolute output	<code>TotalAbsPow</code>
Visualization	Bar chart + Topomap (per-band colorbar)
Main caveat	Relative power depends on selected bands
EEG caveat	Gamma is susceptible to EMG and other artifacts

Feature flow

EEG → MNE Raw → YASA bandpower() → Channel × Band values → Bar / Topomap

This example demonstrates how a YASA feature can be extracted from an MNE EEG object and reused with MNE visualization tools.

References

YASA

Vallat, R., & Walker, M. P. (2021). An open-source, high-performance tool for automated sleep staging. *eLife*, 10, e70092. <https://doi.org/10.7554/eLife.70092>

Official YASA documentation and API reference: `yasa.bandpower` — <https://yasa-sleep.org/generated/yasa.bandpower.html>

MNE-Python / EEGBCI

MNE-Python documentation: EEGBCI dataset loading and EEGBCI run definitions — https://mne.tools/stable/generated/mne.datasets.eegbci.load_data.html

Dataset

EEG Motor Movement/Imagery Dataset (EEGBCI), PhysioNet — <https://physionet.org/content/eegmmidb/1.0.0/>

Scientific reference for EEGBCI

Schalk, G., McFarland, D. J., Hinterberger, T., Birbaumer, N., & Wolpaw, J. R. (2004). BCI2000: A General-Purpose Brain-Computer Interface. *IEEE Transactions on Biomedical Engineering*, 51(6), 1034–1043.

No local file paths are used. The EEG dataset is obtained through MNE's public dataset loader.

Reproducibility / Environment

Required Python packages:

- `marimo`
- `mne`
- `numpy`
- `matplotlib`
- `yasa`

The example uses public data and contains no hard-coded local dataset paths.

Recommended project configuration:

```
[project]
dependencies = [
    "marimo",
    "mne",
    "numpy",
    "matplotlib",
    "yasa",
]
```

The notebook can be launched with:

```
marimo edit <filename>.py
```

or executed as an application with:

```
marimo run <filename>.py
```

Final Checklist

✓ Clear title ✓ Library identified ✓ Method identified ✓ Feature identified ✓ Domain identified ✓ Dataset described ✓ Subject and Run specified ✓ Sampling frequency displayed ✓ Channels described ✓ Preprocessing visible ✓ No hidden transformations ✓ Main API call visible ✓ Important parameters explained ✓ Numerical output displayed ✓ Output shape explained ✓ Relevant visualization ✓ Axes and units identified (bar chart labels; per-band topomap colorbars) ✓ Mathematical interpretation ✓ EEG interpretation with appropriate caution ✓ No unsupported clinical claims ✓ Public dataset ✓ Official documentation identified, with links ✓ Scientific reference identified, with correct citation ✓ Atlas Summary included ✓ No local paths